MIL-C-24166A(SHIPS)
7 December 1971
SUPERSEDING
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25 August 1965
(See 6.5)

MILITARY SPECIFICATION

COOLER, DRINKING WATER, MECHANICALLY REFRIGERATED, SELF-CONTAINED, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers mechanically refrigerated drinking water coolers for Naval shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766 - Steel, Plates, Sheets, and Strip - Corrosion Resisting.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests (H.I.) High-Impact; Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C., and Associated Switching Devices, Naval Shipboard.
MIL-M-9868 - Microfilming of Engineering Documents, 35MM, Requirements for.
MIL-M-9868/1 - Microfilming of Engineering Documents, 35MM, For Naval Ship Systems.
MIL-C-9877 - Cards, Aperture.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements For.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts; Packaging and Packing of.
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-M-38761 - Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction System, General Requirements For, Preparation of.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-804 - Formats and Coding of Aperture, Copy and Tabulating Cards for Engineering Data Micro-Reproduction System.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
MG1-18.076-.091 - Motors for Hermetic Refrigeration Compressors.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B-88 - Specification for Seamless Copper Water Tube.
B-280 - Specification for Seamless Copper Tube for Refrigeration Field Service.
C-177 - Method of Test for Thermal Conductivity of Materials by Means of the Guarded Hot Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B-18.6.4 - Slotted and Recessed Head Tapping Screws and Metallic Drive Screws.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

UNDERWRITERS' LABORATORIES (UL)
STANDARDS FOR SAFETY
399-Water Coolers, Drinking.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article samples. Prior to beginning production, a sample shall be tested as specified in 4.2 (see 6.3).

3.2 Materials (general requirements).

3.2.1 All materials shall be free from any defects that might affect the serviceability or appearance of the finished product. Materials shall be in conformance with the requirements specified herein. Consideration will be given to alternate materials which will give equal or better service.

3.2.1.1 Materials which are capable of producing dangerous gases or other harmful effects under conditions (including fire) encountered in Naval shipboard service shall not be used without approval of the command or agency concerned. Magnesium and its alloys shall not be used.

3.2.2 Protective coatings.

3.2.2.1 Protection against electrolytic or galvanic corrosion. Water coolers shall be constructed with careful attention to protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wetted in service shall be protected against electrolytic or galvanic corrosion by proper selections of materials, plating, isolation, insulation, paint, area relationship, gaskets or other means.

3.2.2.2 Corrosion protection. Corrosion resisting steel, copper and brass referenced herein are considered satisfactory corrosion resisting materials. Corrosion resisting

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steel, when fabricated by any method that tends to reduce corrosion resisting properties, shall be normalized to restore these properties before assembled in any unit. Where corrosion resisting steel is specified, it shall conform to class 302 or 304 of QQ-S-766 and where exposed in final assembly shall have a number 4 or 6 finish.

3.2.2.2.1 Parts other than corrosion resisting materials shall be protected against corrosion after fabrication with chemicals, electrolytic processes, plating, or suitable paints and enamels. The following methods, while not restrictive, are acceptable corrosion protection methods when properly applied:

(a) Hot dipped galvanized.
(b) Hot phosphate process with rust inhibiting paint.
(c) Electroplated cadmium followed by dichromate, or hot phosphate treatment or water dip lacquer.
(d) Caustic dip, sandblast process, "anodized" coated with phenolic resin baked on, or covered with lacquer for aluminum.

3.2.2.2.2 All screws exposed in final assembly shall be corrosion resisting steel. All other bolts, nuts, pins, screws and such other fastenings shall be of the same material being joined or supported or shall be cathodic to the materials being joined or supported. Self-tapping spaced thread screws of the thread forming type (ANSI B-18.6.4, types A, B, and BP) will not be permitted.

3.2.2.2.3 Painting. All external unmachined surfaces of metal parts, except corrosion resisting steel, copper and brass, shall be thoroughly cleaned and coated with paint in accordance with the following or equivalent materials:

(a) One coat of pretreatment coating.
(b) One coat of zinc chromate primer or one coat of red lead primer.
(c) Finish coat paints or enamels shall be of the baked on type.

3.2.3 Thermal insulation. Thermal insulation materials, other than pipe insulation, shall have a thermal conductivity not greater than 0.30 British thermal units (B.T.U.) per hour per square foot per inch thickness per degree Fahrenheit (F.) temperature difference at a mean temperature of 50°F. when tested in accordance with the "guarded hot plate method" specified in ASTM C-177. Insulation shall be applied in such a manner as to preclude the possibility of settling in service. The insulation shall be provided with a vapor barrier and designed to prevent accumulation of moisture and possible saturation of insulation. All insulation spaces shall be filled so as to leave no voids.

3.2.4 Interchangeability. Similar parts, including repair parts, or corresponding apparatus furnished on the same contract or order or built to the same drawings shall be strictly interchangeable without the necessity of further machining or hand fitting of any kind.

3.2.5 Reliability assurance.

3.2.5.1 The principle of maximum reliability is paramount and no compromise of this principle shall be made with any other basic requirement of design. Where wear or corrosion is unavoidable, the parts subjected to such wear or corrosion shall be of the best material for the purpose in order to reduce these detrimental effects to a minimum.

3.2.5.2 The designer should take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. Human engineering should be considered in design to preclude or minimize possibility of failure through improper operation or poor maintenance.

3.2.5.3 The supplier shall establish and maintain an effective reliability assurance program including the items specified in 3.2.5.3.1 through 3.2.5.3.4.

3.2.5.3.1 Design reviews. The reliability assurance program shall include provisions for the reliability review and evaluation of designs as an integral part of the supplier's engineering design procedures. Design or engineering changes occurring during development or production shall be subjected to comparable review procedures.

3.2.5.3.2 Production control and monitoring. The reliability assurance program shall provide an economical and effective system of production control and monitoring to assure that reliability achieved in design is maintained during production (see 6.1).

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3.2.5.3.3 Subsupplier and vendor reliability. The reliability assurance program shall include provisions to assure subsupplier and vendor selection and performance consistent with the reliability requirements of the contract and applicable portions of this specification.

3.2.5.3.4 Failure reporting, analysis and feedback. The reliability assurance program shall incorporate a formalized system for recording, collecting, and analyzing all failures that occur during all testing, installation and operation through the tenure of the contract. Analysis shall be fed back to supplier's engineering, management, and production activities on a timely basis. Failure reports received from using activity shall be integrated into this program for trouble analysis and for experience considerations for problem solving and design improvement for future.

3.3 General assembly. The drinking water cooler shall consist of a condensing unit, water cooling unit, controls, interconnecting water and refrigerant piping and miscellaneous accessories. The various assemblies and components shall be supported by an integral frame. The drinking water cooler shall not exceed 17 inches square and 44 inches in height from the top rim of the basin to the base. Wood shall not be used in the equipment construction. The drinking water cooler shall be factory assembled, complete, self-contained with the refrigerating system dehydrated and charged with the proper operating quantities of refrigerant and oil. The unit shall be ready for operation after removal of shipping protection, opening of valves, adjustment of belts and connection to services.

3.3.1 Cabinet. The integral frame shall be constructed of welded structural members. Four 1/2 inch diameter bolt holes shall be provided in the structural members forming the base for bolting the unit to a deck. The holes shall be located in each corner of the cooler spaced 12-1/4 inches center to center. The frame assembly shall be treated for corrosion resistance after assembly. The base members or any members forming legs for securing the unit to a deck shall be treated for corrosion resistance by galvanizing.

3.3.1.1 The cabinet top shall be fabricated from corrosion resistant steel and be drawn in one piece to form a basin. The sides of the top shall overlap the front, sides and back of the cabinet for water shed. The cabinet top shall have a drain opening with a strainer in the basin and shall be provided with openings to permit the mounting and connecting of a bubbler valve and pitcher filler. The basin strainer openings shall be not greater than a 1/4 inch diameter hole in area. The front and sides of the unit shall be fabricated from corrosion resistant steel not less than 19 gage (AWG) and may be formed either as a wrapper or as individual panels. All gas or electric arc welded joints, visible in final assembly, shall be ground or polished to match the surfaces adjacent to the welds.

3.3.2 Valves, controls, and equipment subject to service and repair shall be readily accessible for servicing through a removable front panel. Where panels are provided for access to equipment, machine screws may be used for fastening.

3.3.3 Adequate ventilation shall be provided for condensing unit equipment. Where ventilation is provided in the front or sides of cabinet, the opening shall be louvered type. Slotted openings shall be arranged for water shed. Louvered openings for ventilation shall shall not exceed 1/2 inch in clear width and shall not have sharp or ragged edges. Electrical devices shall be located so as to be protected or shall be shielded from water spillage or drippage.

3.4 General design. The equipment shall be designed for operation under shipboard conditions of vibration, pitch, roll and shock. The design and construction of the equipment shall conform to the requirements of UL-399. Coolers shall be for operation with 115 volt, single phase, 60 cycle, alternating current (a.c.) or 115 volt direct current (d.c.) power supplies as specified (see 6.1).

3.4.1 Shock and vibration.

3.4.1.1 The design of all complete drinking water coolers shall be such that they are capable of passing the high impact shock tests specified in MIL-S-901 for grade B, class I equipment.

3.4.1.2 The general information given in MIL-S-901 should be carefully considered in design of equipment. This information is also used by the drawing review activity as a factor in approval of drawings required for production equipment.

3.4.1.3 Shock mounts shall not be used unless approval of the mount and its proposed application is given by the command or agency concerned.

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3.4.1.4 The design of equipment and assembly shall be such that damage will not occur or that malfunction be caused either by internally excited vibrations or by the environmental vibrations specified in MIL-STD-167 for frequencies up to and including 33 cycles per second (c.p.s.).

3.4.2 Inclination. The equipment shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other (see 4.2.2).

3.5 Condensing unit. The condensing unit shall be designed for operation with dichlorodifluoromethane (R-12) refrigerant and shall consist of a compressor, condenser, receiver (if required), fan, motor and accessories mounted on a fabricated steel frame. A capillary tube shall not be used as the refrigerant expansion device when an open type compressor is employed.

3.6 Compressor. The compressor shall be of the hermetic type for a.c. or open type for d.c. Provision shall be made for adequate lubrication of all rubbing and wearing surfaces, including operation under shipboard conditions of inclination and rolling.

3.6.1 The open compressor shall be of the vertical, single acting, reciprocating type. The shaft seal shall be replaceable in its entirety without the necessity of replacing or refinishing the crankshaft. The suction and discharge shut-off valve connections to the compressor shall be of the flanged type.

3.6.2 The hermetic type compressor shall consist of a compressor and a motor enclosed in a gas-tight shell.

3.6.3 The capacity rating of the compressor, based on operation in an ambient temperature of 90°F., shall be not less than 2500 B.T.U. per hour under the conditions of 40°F. evaporating temperature.

3.7 Condenser. The condenser shall be forced air cooled and constructed of finned tubes. The tubes shall be not greater than five rows deep. Fins shall be soldered, welded, or mechanically bonded to the tubes and shall be not less than 0.010 inch thick and spaced not more than six fins per inch. The condenser shall be treated for corrosion resistance after fabrication. The fan shall be located as nearly as is practicable to the center of the condenser face and where necessary for uniform air flow, a metal fan shroud shall be provided. The condenser shall have sufficient cooling surface area to permit compressor operation in an ambient temperature of 120°F. with a maximum discharge pressure of not more than 215 pounds per square inch gage (p.s.i.g.) and an evaporating temperature of 40°F.

3.8 Liquid receiver. The liquid receiver, when furnished, shall be such that it has an internal volume at least 25 percent greater than the liquid volume of the complete refrigerant charge. A liquid receiver shall be furnished when an open type compressor is used and may be used in a hermetic system. The receiver shall contain a fusible plug for protection against excessive temperatures. The plug shall melt at a temperature not lower than 200°F. and not higher than 300°F.

3.9 Refrigerant piping. All fittings and joints, where practical, shall be brazed with silver brazing alloy or phosphorous copper. Connections shall be kept at a minimum and, where tubing joints are necessary, couplings shall be used. Piping adjacent to frame members or to other components or parts shall be rigidly supported to prevent movement and rubbing as a result of environmental vibration.

3.9.1 Piping shall conform to ASTM B-280. All fittings shall be of wrought copper or forged brass.

3.9.1.1 The suction line to compressor within machinery compartment shall be insulated to prevent condensation.

3.9.1.2 A dehydrator shall be provided in the refrigerant circuit. Each dehydrator shall be equipped with an auxiliary screen or other protective means at the dehydrator outlet to prevent passage of dehydrating agent in the event of rupture of the cartridge screen outlet. The desiccant shall be silica gel or activated alumina or a type satisfactory to the command or agency concerned.

3.9.1.3 A heat-interchanger shall be provided in the suction line. Bonding of the liquid refrigerant line to the suction line to provide adequate liquid subcooling is considered satisfactory as a heat-interchanger.

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3.9.2 Piping systems shall withstand the pressure and leak test specified in 4.5.1.

3.10 Cooling unit. The cooling unit shall consist of a water storage tank with external refrigerant coil and insulation all housed within a cabinet. The water storage tank shall be located within the cabinet so that there is at least 2-1/2 inches of insulation between the tank and the sides, top or bottom of the cabinet. The water storage tank shall be rigidly supported within the cabinet. Water supply and drain piping shall be arranged so as not to come in contact with the refrigerant piping or the water storage tank. The cooling unit cabinet shall be completely filled with a granulated cork or similar insulation.

3.10.1 Water storage tank. The water storage tank shall be of nonferrous metal. The internal surfaces of the tank shall be coated with tin, nickel or chromium. The coating may be either by plating or dipping. The tank shall be constructed of two drawn or formed sections requiring only one seam for assembly. The seam shall be silver brazed with minimum overlap of 3/8 inch. Water outlet connections shall be so arranged that sediment may readily be flushed out. Means shall be provided to avoid short circuiting of warm incoming water to the outlet. One or more wells shall be provided in the top of the tank for insertion of the operating and low temperature safety thermostatic switch thermal bulbs. Where a single well is used, the operating bulb shall be located in the bottom of the wall and the safety thermostat near the top or in the area where the lowest water temperature will occur. The water storage tank and water piping shall withstand the pressure and leak test specified in 4.5.2 and 4.5.2.1.

3.10.1.1 Refrigerant coil. The refrigerant coil shall be nonferrous, external to the water storage tank and continuously bonded to it. The use of additional refrigerant to water cooling by means of tube in tube or tube on tube exchanger will not be permitted. Sufficient coil surface shall be provided to permit operation in accordance with the requirements of 3.12.

3.10.2 Water and drain piping. The piping and fittings shall be made of brass or copper. Copper tubing, where used, shall conform to ASTM B-88 type "L". Internal water piping connections shall be silver brazed joints, using silver brazing alloys, or phosphorous copper. Fittings for external connections shall be secured to the frame or to a bracket secured to the frame in such a manner that when connections are made to external water and drain lines, no strain will be placed upon the internal piping. Offsets shall be incorporated within the cooler on all lines to relieve stress due to vibration. Piping adjacent to frame members or other component parts shall be rigidly supported to prevent movement and rubbing as a result of environmental vibration. Fresh water supply, waste and tank drain connections shall be provided. Water supply and tank drain connections shall be female 3/8 inch national taper pipe (thread) (NPT) or union with solder fittings provided with plugs. The waste connection shall be 1-1/4 inch female NPT or union with solder fittings provided with plug. The pitcher filler connections shall be male 3/8 inch NPT and provided with a brass, chrome plated, hexagon pipe cap. Water lines within cabinet shall be insulated.

3.10.3 Valves. The bubbler and pitcher filler valves shall be self-closing and made of brass throughout. Exposed parts shall be chrome plated. The valves shall be constructed for rough usage.

3.10.3.1 The bubbler shall be sanitary and shall have a slanting jet, protected by a guard and designed to prevent contact with the mouth. The guard shall be designed to prevent foreign matter from dropping vertically into the orifice of the bubbler. The orifice of the bubbler shall be located at a point not less than 3/4 inch above the overflow level of the top. The bubbler and guard shall be made of brass, chrome plated.

3.10.3.2 The pitcher filler, when required (see 6.1), shall be of brass, chrome plated and shall be suitable for filling a pitcher 11 inches high. Waste water from the pitcher filler shall drain to the basin. The pitcher filler shall incorporate a self-closing valve and have a female 3/8 inch NPT connection for connecting to the drinking water cooler. The pitcher filler, when required, shall be packed separately for installation by the installer.

3.10.3.3 Stream regulator. An automatic stream regulator of brass material shall be provided in the bubbler line and shall operate satisfactorily on fresh water system pressure ranging from 20 to 70 p.s.i. The regulator shall maintain a bubbler-stream height of not less than 1 inch nor more than 3 inches above the top of the bubbler guard for any one setting without readjustment, and shall be readily accessible for adjustment or repairs.

3.10.4 A pre-cooler, where provided, shall be constructed of nonferrous seamless tubing. Both inlet and drain water passages shall be fully enclosed with two walls of metal separating the incoming drinking water from the drain water. There shall be no joints in

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the area of contact of the drain and supply water tubing. Surfaces in contact with the drain water shall be smooth and there shall not be any internal restrictions or passages that would reduce the free opening to less than 5/8 inch diameter. The pre-cooler shall be securely mounted within the cooling unit housing and shall be located a minimum of 1 inch from the water storage tank.

3.11 Temperature control. The automatic control of the compressor operation shall be accomplished by means of a temperature control switch. The switch shall be arranged to start and stop the compressor motor when the temperature of the water in the storage tank rises above or falls below predetermined values. The temperature control switch shall be set to maintain design water temperature. The switch shall include a stop to prevent setting the switch to open at a temperature lower than 35°F. A low temperature safety thermostat shall be installed. The safety thermostat shall be arranged to stop the compressor when the temperature of the water in the storage tank is 35°F. The switch shall be a wide range type or a type that requires manual resetting to start the compressor. Operation of the switch contacts under the shock test is permissible.

3.12 Capacity rating. The cooler shall have a minimum capacity rate of 7-1/2 gallons per hour based on the following conditions:

Ambient temperature	90°F.
Water in	80°F.
Water out	50°F.
Evaporator refrigerant temperature (minimum)	32°F.

3.12.1 Peak draw capacity. The outlet temperature shall not exceed 55°F. during a 15-minute period while drawing 2.75 gallons of water from the cooler. Under this condition, outlet water temperature shall be 45° ±2°F. at the beginning of the 15-minute period (see 4.2.3.2).

3.12.2 Maximum operating conditions. The coolers shall start and operate satisfactorily in an ambient temperature of 120°F. with inlet water temperature of 90°F. Under these conditions at rated capacity the water outlet temperature shall not exceed 65°F. (see 4.2.3.3).

3.12.3 Overload condition. The unit shall operate satisfactorily in an ambient temperature of 120°F., inlet water of 90°F. and water drawn at the rate of 15 gallons per hour (see 4.2.3.4).

3.13 Basin drain. The cooler shall free drain the cabinet top basin when the cooler is tilted 15 degrees in any direction (see 4.2.2).

3.14 Nozzle flow. With the fresh water supply at 35 to 40 p.s.i.g., the water regulator and bubbler shall deliver water at the minimum rate of 1/2 gallon per minute. With a water supply line pressure range of 20 to 75 p.s.i.g., the water regulator device shall not permit the flow rate through the bubbler to deviate by more than 20 percent from the flow rate at supply line pressure of 35 to 40 p.s.i.g.

3.15 Compressor motors. Alternating current motors shall conform to the requirements of NEMA MG1-18.076-.091. Direct current motors shall comply with MIL-M-17556. Compressor motors shall be not less than 1/4 horsepower. The motor shall be constant speed, continuous duty and shall be protected against overload by a thermal protector built into the motor or mounted on the motor compressor unit. The temperature rise of the hermetically sealed motor shall not exceed 70°C. In addition to the above, direct current compressor motor shall conform to the following:

Service - C.
Enclosure - Dripproof.
Bearings - Ball.
Rpm - 1800 nominal.
Winding - Shunt (or stabilized shunt) constant speed.
Ambient temperature - 50°C.
Insulation - Class A.
Under maximum operating conditions (see 3.12.2) the motor temperature rise shall be not more than 15°C. above corresponding temperature rises permitted in the motor specification.

3.15.1 Motor controller direct current unit. The d.c. unit shall be provided with a manual motor controller conforming to MIL-C-2212. The controller shall provide overload protection.

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3.15.2 Fan motor. When a separate motor is used to drive the condenser cooling fan, the fan motor enclosure shall be dripproof. The motor shall be of a type that has a sealed or enclosed lubricant system and does not require bearing lubricant to be changed or supplemented periodically.

3.15.3 Motor starting relays. Motor starting relays, used in lieu of centrifugal switches for disconnecting the starting winding of single-phase a.c. motors, shall be mechanically shockproof (as mounted on the refrigeration equipment when tested in accordance with MIL-S-901) and dripproof in accordance with MIL-E-2036. Rubber or other shock-mounts which are subject to deterioration in service shall not be used for mounting the relays.

3.15.4 Wiring. The wiring between the electric devices (motors, motor controllers, control devices) shall conform to the requirements of UL-399. The wiring shall be complete and the equipment shall be ready for connection of the incoming power cable at installation. A terminal block conforming to MIL-E-917 for three-conductor power cable shall be provided for the incoming power cable.

3.15.5 Insulation resistance. The insulation resistance of the electrical circuit shall be not less than one megohm (see 4.5.5).

3.15.6 Dielectric strength. The equipment shall be capable of withstanding for a period of 1 minute a dielectric a.c. root mean square (r.m.s.) test voltage of 1900 volts for the a.c. unit and 1500 volts for the d.c. unit (see 4.2.3.8).

3.16 Automatic refrigeration control devices shall be readily accessible for adjustment and servicing. Where thermostatic expansion valves are used, they shall be of the pressure limiting type. Where the compressor is a low starting torque type, the expansion valve shall have a bleed port for system pressure equalization.

3.17 Identification plates. Each water cooler shall be provided with an identification plate. The type of marking for the letters shall be Gothic capitals, and numerals and other characters shall be of similar appearance. The size of the letters, numerals and other characters shall be as necessary for easy readability. The plates may be etched, engraved, or cast using either nickel copper alloy brass, corrosion resisting steel, or aluminum alloy, or may be of the anodized aluminum photographic process.

3.17.1 Anodized aluminum photographic process. Anodized aluminum plates shall have the photosensitive silver compounds imbedded within the oxide layer. After photographic processing the image shall be sealed in the oxide layer by a chemical treatment of the layer. Stamping of additional marking information shall not be permitted.

3.17.2 The identification plate shall contain the following information:

(a) Item name.
(b) Model and size (when applicable).
(c) Federal stock number.
(d) Component identification number (CID) (nine digit).
(e) Supplier's number.
(f) Manufacturer's name and address.

3.17.3 The identification plate shall be located on the front of the cooler.

3.18 Repair parts. Repair parts shall be furnished in accordance with MIL-P-15137 (see 6.2). The number of sets shall be as specified (see 6.1). The repair parts shall consist of those specified in table I, as applicable, and shall constitute one set.

3.19 Drawings. Drawings delineating the equipment shall be furnished. Drawings shall be type II, III or IV of MIL-D-1000/2, as applicable. The drawings shall be for intended use categories A, B, G and H as delineated in MIL-D-1000. Separate drawings need not be furnished for each category. Parts lists, bill of material lists, etc. shall be included on equipment drawings. Certification data drawings should include diagrams and drawing lists. Unless otherwise specified (see 6.1), preliminary drawings for approval and final approved drawings shall be furnished on each procurement.

3.19.1 Assembly drawings. Assembly drawings type II or IV shall be provided for the complete drinking water cooler and the following components, as applicable:

(a) Compressor.
(b) Compressor motor.

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(c) Condenser.
(d) Condenser fan assembly.
(e) Receiver.
(f) Evaporator assembly.
(g) Controls.

Table I - Onboard repair parts.

Item	For quantity of onboard repair parts required per ship, multiply the allowance factor in the applicable component (refrigerator, etc.) column by the quantity installed in one component and round the product to the nearest whole number.							
	Number of identical components per ship							
	1	2	3	4	5-8	9-20	21-50	51-100
Compressor repair parts (open compressor):								
Valve plate assembly complete with suction and discharge valves	0.20	0.40	0.60	0.60	0.80	1.5	3.00	3.50
Set of suction and discharge valve disks	.20	.40	.60	.60	.80	1.5	3.00	3.50
Set of gaskets of each size and type	.20	.40	.60	.60	.80	1.5	3.00	3.50
Shaft seal assembly complete	.20	.40	.60	.60	.80	1.5	3.00	3.50
Set of belts	.25	.50	.75	.75	1.0	2.5	3.50	4.00
Miscellaneous repair parts:								
Valve of each type and size used	.33	.66	1.00	1.00	2.00	3.00	4.00	5.00
Control switch of each type and size used	.33	.66	1.00	1.00	2.00	3.00	4.00	5.00
Liquid strainer basket	.25	.50	0.75	0.75	1.00	2.50	3.50	4.00
Capillary tube assembly (where used)	.25	.50	.75	.75	1.00	2.50	3.50	4.00
Dehydrator	.33	.66	1.00	1.00	2.00	3.00	4.00	5.00
Bubbler valve and stream regulator valve	.33	.66	1.00	1.00	2.00	3.00	4.00	5.00
Electrical repair parts:								
Hermetic compressor motor:								
Relay	.33	.66	1.0	1.0	2.00	3.00	4.00	5.00
Overload	.25	.50	0.75	0.75	1.00	2.50	3.50	4.00
Capacitor	.33	.66	1.00	1.00	2.00	3.00	4.00	5.00
Condenser fan motor (hermetic compressor):								
Motor with fan complete	.10	.20	0.30	0.40	0.50	0.60	1.50	2.50
Direct current compressor motor and motor controller	See MIL-M-17556 and MIL-C-2212, as applicable, except no special tools are required.							

3.19.1.1 Assembly drawings shall show the following information:

(a) Outline dimensions.
(b) Mounting, attachment and connection dimensions.
(c) Methods and sizes of fastenings or connections.
(d) Weight and center of gravity of each complete unit.
(e) Clearances required for operation and maintenance.
(f) All data necessary to permit installation of the equipment without the supplier's assistance.

The drawings shall illustrate the design, construction, operation (or function) and identity of parts. Section or partial views either on the assembly or detail drawings shall show attachment and fastenings of cabinet hardware, supports, evaporator mountings and support and stiffening of cabinet or panels where provided. Subassembly drawings shall be furnished where assembly drawings do not adequately describe and identify subassembly parts and components. The drawings shall be sufficiently complete to reflect compliance with equipment specification requirements. Where first article tests are required, the tests should be identified and test approval authority and date should be noted on the drawing.

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3.19.2 Certification data sheets. Certification data sheets, type III drawing(s), shall be provided. The drawing(s) shall include a reference drawing list of all the assembly, subassembly and detail drawings of the equipment and components being furnished on the contract or order. Diagrams shall also be provided. The diagrams shall show by symbolic representation all fluid piping, pipe sizes, electrical interconnections, components, accessories and controls plus connections required by others for operation. For example, the refrigerant diagram will include the associated condensing unit, controls, valves, piping accessories and evaporator as applicable. The electrical wiring diagram will include and show circuits between motors, controllers, switches and accessories as applicable for understanding the operation sequence. All symbols used for equipment in diagram drawings shall be given a piece number and identified in the list of material with the following information:

(a) Piece number.
(b) Quantity required.
(c) Descriptive name.
(d) Manufacturer's name.
(e) Manufacturer's identifying model or part number.
(f) Weight.
(g) Characteristics.

3.19.2.1 Characteristic data. The characteristics shall include ordering information necessary to limit or supplement data on referenced drawings or the manufacturer's model or part number. Complete characteristics for the refrigerator as well as for all major parts or components including the compressor, condenser, receiver, cooling coils, fan and fan motor, switches, accessories and expansion valve as applicable shall be furnished.

3.19.3 Detail drawings. Detail drawings shall be furnished for all parts and subassemblies necessary for evaluation of the equipment and all parts required for maintenance and overhaul of the equipment. Subassemblies whose parts cannot be procured or serviced individually should be shown as a single part. Multi-detail drawings are preferred but mono-detail drawings may be furnished. Drawings are not required for those parts which are in common commercial use and can be referenced to commercial standards.

3.19.4 Electrical component drawings. In addition to the drawing requirements specified herein, the content of electrical component drawings shall conform to the requirements of the applicable component specification.

3.19.5 Microfilm.

3.19.5.1 Microfilm mounted in aperture cards shall be as final drawings, supplied with coverage of all drawings applicable to all items of equipment. Microfilm shall be type I, class 1 and type II, class 2 in accordance with MIL-M-9868. One sheet per frame format shall apply. A set of cards shall consist of that number of cards necessary for the required coverage.

3.19.5.2 Tabulating index cards shall be supplied for aperture cards. A set of tabulating index cards shall consist of an index card for each aperture card in a set of microfilm aperture cards as defined herein.

3.19.5.3 Selection and preparation of aperture and tabulating index cards shall be in accordance with MIL-STD-804, MIL-M-38761, and MIL-C-9877.

3.19.5.4 Dual purpose engineering document card (DD form 1562). Card code "T", lower legends of MIL-STD-804 shall be obtained commercially and used in the preparation of all aperture and tabulating index cards.

3.19.5.4.1 Aperture and tabulating index cards shall be punched and interpreted in accordance with MIL-STD-804 and the following:

(a) Card columns 31-32; punch the frame number of a multiframe drawing. A single frame drawing shall have frame number '1' entered in right hand column. This information shall be interpreted using interpreter bar locations 29U30.
(b) Card columns 33-34; punch the number of frames required to completely photograph the drawing. The number of frame digit shall be entered in the right hand column. This information shall be interpreted using interpreter bar locations 31U32.
(c) Card columns 35-36 shall be left blank.
(d) Card columns 48-49; insert control activity symbol "HR". This information shall be interpreted using interpreter bar locations 51U52.
(e) Card columns 53-80; punch the drawing title in the tabulating index card only. This information shall be interpreted on both index and aperture cards using interpreter bar locations 12L39.

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3.19.5.5 Security requirements. Aperture cards containing classified drawings shall be identified and marked in accordance with MIL-STD-804.

3.19.5.6 Responsibility to update sets of microfilm aperture cards and tabulating index cards supplied to the Government. Microfilm mounted in aperture cards and tabulating index cards shall be supplied for drawings generated or revised during the time period between initial submittal of card sets of expiration of the supplier's responsibility for last ship set of equipment.

3.20 Technical manuals. Technical manuals shall conform to type I of MIL-M-15071. Photo views of the equipment shall be included as a part of the general description. A section shall be provided containing reduced copies of all drawings required to amplify or illustrate the text including diagram and assembly drawings. Exploded views shall be required for open or serviceable hermetic compressors. The quantity of technical manuals to be furnished shall be as specified (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.1 and 6.2).

4.2 First article inspection. First article inspection shall be conducted on a unit of each power supply and shall consist of the tests specified in 4.2.1 through 4.2.3.8. The unit that has satisfactorily passed the tests may be used as one of the production units after correction of any defects. The supplier may acquire all materials required for use on the contract prior to approval of the unit being tested. Any final assembly work performed by the supplier prior to approval of the unit shall be at the supplier's risk. Where first article tests, as required by this specification, have previously been conducted and equipment offered is the same and manufactured to the same Naval Ship Engineering Center (NAVSEC) approved drawings as that previously tested, accepted, and proven satisfactory, a test report of prior tests conducted or certification of prior approval will be acceptable in lieu of retesting.

4.2.1 Shock tests.

4.2.1.1 One complete cooler of each design shall be subject to the high impact shock tests specified in MIL-S-901. The equipment shall be tested in an operating condition while energized and pressurized.

4.2.1.2 Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory or Government laboratory suitably equipped to conduct tests. Where shock tests are conducted at a Government laboratory, copies of the applicable approved drawings shall accompany the units.

4.2.1.3 Equipment previously shock tested and accepted need not be retested where evidence can be submitted to the command or agency concerned that the original item was subjected to shock tests and successfully passed and that the design of the proposed equipment, its intended service and shipboard installation are such as to result in a equal or better degree of shock resistance.

4.2.1.4 Equipment shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part or otherwise creating a hazard to personnel or other equipment.
(c) Significant effect on performance.

4.2.1.5 Equipment which has been subjected to the high impact shock test and has failed to conform to the requirements will not be acceptable.

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4.2.1.6 Equipment which has been subject to the high impact shock test and has successfully passed this test shall be considered acceptable for such service as the command or agency concerned may authorize. Such equipment shall be reconditioned by the manufacturer as follows:

(a) Correct minor deformations including alignment.
(b) Each part shall be carefully examined by the manufacturer and any part which he considers substandard shall be replaced.

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high impact shock test but which are in excess of the design dimensional tolerance specified on the applicable drawings.

4.2.2 Inclination test. The unit shall be inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other and operated at least 1 hour in each plane with no abnormal variations in temperature.

4.2.3 The unit shall be subjected to the tests specified in 4.2.3.1 through 4.2.3.8. Observations shall be taken at not less than 20-minute intervals: ambient temperature, inlet water, outlet water, flow rate of outlet water voltage at service connections, refrigerant suction and discharge pressures shall be recorded. Where a pre-cooler is provided, 60 percent of the water drawn may be used as spillage through the pre-cooler during testing under 4.2.3.1 through 4.2.3.4.

4.2.3.1 Rated capacity. The unit shall be operated under the conditions specified in 3.12 and with the control set to maintain a temperature of the water of approximately 45°F. Water shall be drawn at a rate at which the temperature of the outlet water will not exceed 50°F. The test shall continue until the water rate, head pressure, and outlet water temperature are constant. In no case shall the length of time for this test be less than 3 hours. The cooling capacity of the water cooler during this test shall be not less than the minimum capacity required.

4.2.3.2 Peak draw test. The water cooler shall be installed in a 90°F. ambient temperature space, and the thermostat set to maintain a temperature of 45°F. The test shall start when the condensing unit cuts out on one of its normal no-load cycles. With inlet water temperature of 80°F. and outlet water not less than 45°F., a total of not less than 2.75 gallons of water shall be withdrawn for a 15-minute period. Under these conditions the final outlet water temperature shall not exceed 50°F.

4.2.3.3 Maximum operating condition. The test shall be identical to the rated capacity test (see 4.2.3.1) except that the ambient temperature shall be 120°F. Under this condition, the outlet water temperature shall not exceed 65°F. Immediately following this test and without changing any conditions, the electric power to the cooler shall be turned off twice for a period of 5 minutes each. The electricity shall be turned on again at the end of each off-period and the compressor shall start and operate satisfactorily for at least 5 minutes without tripping the compressor motor overload device.

4.2.3.4 Overload test. The drinking water cooler shall be installed in a 120°F. ambient temperature space with inlet water at 90°F., water drawn at the rate of 15 gallons per hour. The cooler shall operate without tripping the compressor-motor overload device or without breakdown. The test shall be continued until steady conditions have been observed for at least 4 hours. The motor temperature rise shall be not greater than that specified in 3.15. The temperature rise of the windings in the compressor motor shall be measured and computed by the resistance method.

4.2.3.5 Automatic temperature control. The control shall be set at its lowest setting. Before beginning the test, the compressor shall be operated until at least two thermostat cycles occur. The unit shall be operated for several cycles while checking water temperatures at intervals. At least two readings shall be taken immediately after the thermostat has stopped the compressor and at least one immediately after the thermostat has started the compressor. The minimum temperature of the water sample shall be not lower than 35°F. The maximum water temperature shall be not higher than 55°F. At the conclusion of the tests, the thermostat shall be reset to maintain 45°F.

4.2.3.6 Water flow. The rate of water flow shall be measured at water supply line pressures of 20, 40 and 70 p.s.i.g. to determine compliance with 3.10.3.3 and 3.14.

4.2.3.7 Condensation test. The unit shall be tested by operating for 3 hours after establishment of design operating conditions with an ambient temperature of 90°F. and an ambient relative humidity of not less than 50 percent. During the test there shall be no visible evidence of moisture forming on the external cabinet due to condensation or dripping.

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4.2.3.8 Dielectric strength test. With the motor at ambient temperature, the dielectric strength shall be as specified in 3.15.6. The dielectric strength test shall be conducted with the line terminals together and manually closing switches and controls.

4.3 Sampling.

4.3.1 Lot. All coolers of the same design and power supply offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examination. Sample units shall be selected from each lot in accordance with table II and examined as required in 4.4. If any unit is found nonconforming in any requirement, it shall be counted a defective unit, and if the number of such defective units in any sample exceeds the acceptance number shown in table II for that sample, the lot represented by the sample shall be subject to rejection.

Table II - Sampling for examination.

Lot size number of units	Sample size number of units	Defective units	
		Acceptable number	Rejection number
1 to 6	All	-	-
7 to 15	7	0	1
16 to 25	10	0	1
26 to 40	13	0	1
41 to 65	17	1	2
66 to 110	22	1	2
111 to 180	28	2	3
181 to 300	35	2	3
301 to 500	45	3	4

4.4 Examination. Each sample unit selected as specified in 4.3.2 shall be examined for conformance to the requirements of this specification as follows:

(a) The compressor valve seal caps, shall be tight for shipment and all other seal caps in the refrigerant system provided with a gasket and tight against refrigerant leak.
(b) Bolts, nuts, and screws shall be tight; equipment and parts shall be properly fastened and secured.
(c) No parts shall be fractured, split, torn, dented or otherwise damaged such as to effect serviceability.
(d) All cold lines shall be properly insulated (see 3.9.1.1).
(e) The limiting and mounting dimensions shall be as specified.
(f) Electrical wiring shall be properly installed and connected (see 3.15.4).
(g) There shall be no sharp or ragged edges on the sheeting that may be injurious to personnel.
(h) Drinking water cooler shall be drained.
(i) Fittings for the external connections shall be properly secured to the frame (see 3.10.2).
(j) Refrigerant system shall be free of refrigerant leakage.

4.5 Quality conformance tests. Each unit shall be subjected to the tests specified in 4.5.1 through 4.5.5.

4.5.1 Pressure and leakage test. All refrigerant containing parts on the high pressure side of the refrigeration system shall withstand, without damage, an air and gas test of 235 p.s.i.g. The refrigerant containing parts on the low-pressure side of the refrigeration system shall withstand without damage an air and gas test of 140 p.s.i.g. While subjected to the pressures indicated above, the refrigerating system shall be leak tested by means of a halide leak detector and shall be free of evidence of leaks. The halogen leak detector shall have a leak index sensitivity of at least 1/2 ounce per year.

4.5.2 The water system including control valves shall be subjected to a gas leakage test. The system shall be pressurized to 50 p.s.i.g. with 10 percent of R-12 refrigerant as a tracer gas and shall be tested for leaks with a halogen leak detector with a leak index sensitivity of at least 1/2 ounce per year. Any evidence of leakage shall be cause for rejection.

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4.5.2.1 The water system, including control valves, shall be tested at 150 p.s.i. hydrostatic pressure for a period of not less than 10 minutes. Any evidence of leakage or presence of water shall be cause for rejection. The examination for leaks shall be conducted before the insulation is installed.

4.5.3 Operating tests. The units shall be connected to a water supply and operated for a period of at least 1 hour with controls set for specified operating conditions. At the conclusion of this operational test, the entire refrigerant circuit connections, under refrigerant pressure, shall be given a leak test to determine leakage. The detector shall have a leak index sensitivity of at least 1/2 ounce per year. In the event of leakage, all leaks shall be repaired and the performance test shall be repeated.

4.5.3.1 Under equivalent ambient temperature, the electrical power input shall be measured and compared with the input of all other units which have been tested. If any unit requires 7 percent more power than the average of all the acceptable units, it shall be rejected until the cause has been found and corrected.

4.5.4 During and following the pressure and leak testing and operating tests, the water system should be flushed and cleaned to remove all foreign materials. Prior to sealing, the water system should be thoroughly dried.

4.5.5 Insulation resistance test. With the motor at ambient temperature, the insulation resistance shall be as specified in 3.15.5. The measuring of insulation shall be made by connecting the line terminals together, manually closing switches and controls and measuring the resistance between terminals and ground. The insulation resistance shall be measured with a 500 volt insulation resistance meter. The time of test voltage application shall be not less than 60 seconds.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early installation and storage of onboard repair parts.

5.1.1 Water coolers.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, leakage of refrigerant and oil, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, supplier's name and destination.

5.1.2 Onboard repair parts. Repair parts shall be preserved, packaged and packed as follows:

(a) Mechanical repair parts. Mechanical repair parts shall be preserved and packaged in accordance with method I of MIL-P-116. The parts shall be packed level C and marked levels A and C, respectively, in accordance with MIL-E-17555.

(b) Electrical repair parts. Electrical repair parts shall be preserved and packaged level A; packed level C; and marked levels A and C, respectively, in accordance with MIL-E-17555 and MIL-E-16298, as applicable.

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(c) Valves. Valves and fittings, except refrigerant valves, shall be preserved and packaged level A; packed level C; and marked levels A and C, respectively, in accordance with MIL-V-3. Refrigerant valves shall be unit protected and packaged in accordance with MIL-V-3. Refrigerant valves shall be unit protected and packaged in accordance with method 1C of MIL-P-116 without use of contact preservative compounds.

(d) Miscellaneous repair parts. Miscellaneous repair parts such as strainer baskets, capillary tube assemblies, dehydrators, belts and so forth shall be unit protected and packaged in accordance with method III of MIL-P-116, without use of contact preservatives. The parts shall be packed level C and marked levels A and C, respectively, in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging.

5.2.1.1.1 Level A. Each complete unit (see 3.3) shall be unit protected and packaged in accordance with method I of MIL-P-116 and as follows: The water system shall be thoroughly drained and blown out by the application of clean, dry compressed air. All openings including the bubbler and basin drain shall be sealed to prevent entrance of dirt and moisture. Drive belts shall be removed, exposed uncoated ferrous metal surfaces of pulleys and shafts cleaned and coated with primer conforming to TT-P-664. When the primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed. Compressor and motor mounting and shipping bolts, nuts and screws shall be protected from corrosion by the application of preservation compound conforming to grade 4 of MIL-C-16173.

5.2.1.1.2 Level C. Preservation and packaging of each unit (see 3.3) shall be sufficient to afford adequate protection against corrosion, leakage of refrigerant and oil, deterioration and physical damage during shipment from the supply source to the using activity for early installation. This level may conform to the supplier's commercial practice when such meets these requirements.

5.2.1.2 Packing.

5.2.1.2.1 Level A. Each complete unit, packaged as specified in the contract or order, shall be packed in a container conforming to any one of the following specifications at the option of the supplier:

Specification	Classification
PPP-B-601	Overseas type
PPP-B-621	Class 2
MIL-C-104	----------

Wood and wood-cleated boxes shall be provided with waterproof caseliners in accordance with MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Wood or wood-cleated boxes shall not exceed the weight limitations of the applicable box specification. Blocking, bracing, anchoring and cushioning of container contents shall be in accordance with MIL-STD-1186 and the appendix to MIL-C-104. Shipping container closure shall be in accordance with the applicable box or crate specification or appendix thereto.

5.2.1.2.2 Level B. Each complete unit packaged as specified in the contract or order, shall be packed in a container conforming to any one of the following specifications at the option of the supplier:

Specification	Classification
PPP-B-601	Domestic type
PPP-B-621	Class 1
MIL-C-104	----------

Wood or wood-cleated boxes shall not exceed the weight limitations of the applicable box

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specification. Blocking, bracing, anchoring and cushioning of container contents shall be in accordance with MIL-STD-1186 and the appendix to MIL-C-104. Shipping container closure shall be in accordance with the applicable box or crate specification or appendix thereto.

5.2.1.3 Marking. In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.2 Repair parts. Repair parts shall be preserved and packaged level A or C as specified; packed level A, B or C and marked in accordance with MIL-E-17555, MIL-E-16298 and MIL-V-3, as applicable.

5.2.3 Manuals. Manuals shall be packaged, packed and marked for shipment in accordance with MIL-M-15071.

5.2.4 Drawings. Drawings shall be prepared for shipment, for the level specified (see 6.1), in accordance with MIL-D-1000/2.)

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Production control and monitoring system (see 3.2.5.3.2).
(c) Power supply (see 3.4).
(d) Whether pitcher fillers for drinking water coolers are required (see 3.10.3.2).
(e) The number of sets of repair parts to be furnished (see 3.18).
(f) Number of sets of preliminary and final drawings required (see 3.19).
(g) Quantity and destribution of manuals to be furnished (see 3.20).
(h) Quality assurance requirements (see 4.1.1).
(i) Preservation, packaging, packing or marking requirements if other than as specified in 5.1 (see 5.2).
(j) Level of packing required for drawings (see 5.2.4).

6.2 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-P-15137 (see 3.18).
(b) MIL-I-45208 (see 4.1.1).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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